

TIER 1

TINY WHOOP



YOU FLY

Meteor 75 Pro · sub-250 g · safe indoors

TO GET HERE

Gate 1 – all seven Tier 0 badges

ON THIS SHEET

6 badges · all of them to pass Gate 2

2 QUIZ · 2 BENCH · 2 WITNESSED

← FROM TIER 0

your first real flight is checkpoint 1

GATE 2

Full-size unlocked

TIER 2 · PAVO 20 PRO · CINEBOT 35 · 5-INCH →

1

TIER 1 · PT. 1

WITNESSED

Tiny Whoop

Take off, fly a set pattern, and land a whoop in real life.

THE STANDARD

In goggles on the Meteor: take-off, a figure-eight through two gates, an orbit, and a landing on the pad, no wall contact.

WHY First real flight on a sub-250 g drone that can't do much damage.

2

TIER 1 · PT. 2

BENCH

Soldering

Solder a 5-inch ESC and motors; solder and heat-shrink 22 AWG wire.

AT THE BENCH · 6 STEPS

Iron at the right temperature › Tin the pad › ESC pads › Motor wires › 22 AWG splice with heat-shrink › Continuity check with a meter

Photo DM'd to mtucker27@ · a mentor signs it off

WHY Core electronics skill that transfers to robotics.

3

TIER 1 · PT. 3

QUIZ

Electrical Components

Test on choosing motors, ESCs, batteries and props for a given drone.

4 LESSONS, THEN THE QUIZ

1. The six parts

2. Motors, props and thrust **DRILL**

3. Matching motor, ESC, prop and battery

4. Why a 3-inch and a 5-inch differ

10 questions · 80 % passes · no sign-off needed · retake any time

WHY Why a 3-inch and a 5-inch need different parts.

6

TIER 1 · PT. 6

WITNESSED

Freestyle Flight

Powerloop, trippy spin, tiny gaps.

THE STANDARD

A mentor watches in the sim or on the whoop: a powerloop, a trippy spin, and three tiny gaps in one battery.

Practise first in Ground School: Your first flip and roll · The Split-S · The Power Loop · The Matty Flip

WHY Advanced control before flying expensive airframes.

5

TIER 1 · PT. 5

BENCH

Field Diagnostics

Diagnose a drone that won't arm; swap a motor or prop in the field.

AT THE BENCH · 3 STEPS

Prop swap › Motor swap › "Won't arm"

Photo DM'd to mtucker27@ · a mentor signs it off

WHY Keeps a shoot going instead of ending it.

4

TIER 1 · PT. 4

QUIZ

Radio Protocol

Test on ELRS, analog and digital video, control links, and setting up a digital video system to record.

6 LESSONS, THEN THE QUIZ

1. Analog against digital

2. Bands, channels and racing **DRILL**

3. Antennas and diversity

4. Control links: ELRS, Crossfire, binding, failsafe

5. Interference at an event

6. Digital video: goggles, recording and D-log

10 questions · 80 % passes · no sign-off needed · retake any time

WHY Prevents dropouts and interference at events, and gives the editor footage they can use.

FLIGHT BUILD & TECH KNOWLEDGE & SAFETY CREW & LEADERSHIP

Quiz badges: pass at 80 %, no sign-off · Bench and witnessed badges: a mentor who holds the badge signs it off · demonstration, not attendance



FULL BADGE LIST
SCAN